

Altered recognition of antigen is a mechanism of CD8+ T cell tolerance in cancer

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Antigen-specific CD8+ T-cell tolerance, induced by myeloid-derived suppressor cells (MDSCs), is one of the main mechanisms of tumor escape. Using in vivo models, we show here that MDSCs directly disrupt the binding of specific peptide-major histocompatibility complex (pMHC) dimers to CD8-expressing T cells through nitration of tyrosines in a T-cell receptor (TCR)-CD8 complex. This process makes CD8-expressing T cells unable to bind pMHC and to respond to the specific peptide, although they retain their ability to respond to nonspecific stimulation. Nitration of TCR-CD8 is induced by MDSCs through hyperproduction of reactive oxygen species and peroxynitrite during direct cell-cell contact. Molecular modeling suggests specific sites of nitration that might affect the conformational flexibility of TCR-CD8 and its interaction with pMHC. These data identify a previously unknown mechanism of T-cell tolerance in cancer that is also pertinent to many pathological conditions associated with accumulation of MDSCs.